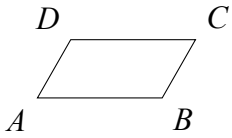
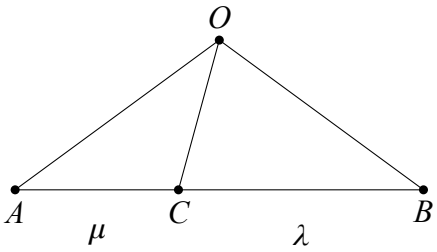
1 Vectors I

1.1 Basic Formulas

	Formula	Note
Magnitude	$\lvert\mathbf{a}\rvert=\sqrt{x^2+y^2+z^2}$	
Unit vector	$\hat{\mathbf{a}}=\dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$	$\lvert\hat{\mathbf{a}}\rvert=1$
Parallel vectors	$\mathbf{a}=k\mathbf{b}$ $\quad\mathbf{a}\times\mathbf{b}=\mathbf{0}$	k scalar constant
Collinear A, B, C	$\overrightarrow{AB}$ // $\overrightarrow{BC}$ (or any two of $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{AC}$)	Try drawing the diagram and take any two vectors, to show that they are parallel.
Parallelogram $ABCD$	$\overrightarrow{AB}=\overrightarrow{DC}$, $\quad\overrightarrow{AD}=\overrightarrow{BC}$	

The Ratio Theorem

Let the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, and suppose that $\lambda AC=\mu CB$. Then the position vector of C is given by the formula

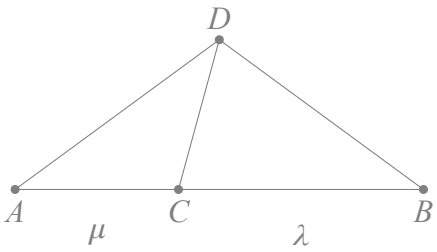


Ratio Theorem

$$\mathbf{c}=\frac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$$

$AC:CB=\mu:\lambda$, so C divides AB with μ beside A and λ beside B.

The same theorem, starting from a point D instead of O:

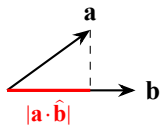
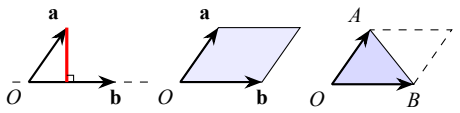


Ratio Theorem

$$\overrightarrow{DC}=\frac{\lambda\overrightarrow{DA}+\mu\overrightarrow{DB}}{\lambda+\mu}$$

1.2 Scalar Product and Vector Product

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}, \qquad \mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}, \qquad \theta=\text{non-reflex } \angle AOB$$

	Scalar Product (Dot)	Vector Product (Cross)
Definition	$\mathbf{a}\cdot\mathbf{b}=a_1b_1+a_2b_2+a_3b_3$	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix} a_2b_3-a_3b_2 \\ -(a_1b_3-a_3b_1) \\ a_1b_2-a_2b_1 \end{pmatrix}$
Result	scalar	vector, $\perp$ to both $\mathbf{a}$ and $\mathbf{b}$
Formula	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$
Angle	$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ $\mathbf{a}\cdot\mathbf{b}>0$ acute, <0 obtuse, $=0$ right angle	$\sin\theta=\dfrac{\lvert\mathbf{a}\times\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ this method only finds the basic angle possible, but usually undesirable
Properties	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$ $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$ $\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$ $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$	$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$ $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$ $\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$ $\mathbf{a}\times\mathbf{a}=\mathbf{0}$
Test	$\mathbf{a}\cdot\mathbf{b}=0\iff\mathbf{a}\perp\mathbf{b}$	$\mathbf{a}\times\mathbf{b}=\mathbf{0}\iff\mathbf{a}\parallel\mathbf{b}$
	(nonzero $\mathbf{a}$, $\mathbf{b}$)	
Applications	Length of projection $=\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ Projection vector $=(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$	Perpendicular distance $=\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ Area of $\triangle OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ Area of parallelogram $=\lvert\mathbf{a}\times\mathbf{b}\rvert$
Geometrical meaning	 $\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ = length of projection of $\mathbf{a}$ onto $\mathbf{b}$	 perpendicular distance from A to line along $\mathbf{b}$ $\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ area of parallelogram, sides $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$ 2× area of $\triangle OAB$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$
Note	$\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law	no cancellation law $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is a vector, but triple cross products are usually not tested

Note

Projection of $\mathbf{a}$ onto $\mathbf{b}\neq$ projection of $\mathbf{b}$ onto $\mathbf{a}$. Perpendicular distance from A to $OB\neq$ from B to OA. Triangles without O as a vertex: $\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$.

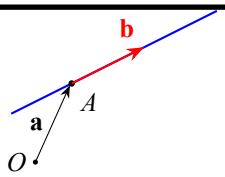
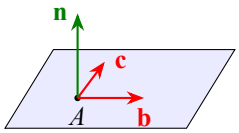
Useful identity

For any vectors $\mathbf{a}$, $\mathbf{b}$: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

$\because\ \mathbf{a}\times\mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$.

2 Vectors II

2.1 Equations of Lines and Planes

	Vector	Cartesian	What you need
Line	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $(\mathbf{r}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ (same line: $\mathbf{r}-\mathbf{a}$ // $\mathbf{b}$)	$\frac{x-a_x}{b_x}=\frac{y-a_y}{b_y}=\frac{z-a_z}{b_z}$ $\mathbf{a}=\begin{pmatrix}a_x\\a_y\\a_z\end{pmatrix}, \mathbf{b}=\begin{pmatrix}b_x\\b_y\\b_z\end{pmatrix}$ if $b_x=0$: $x=a_x$ separately all coefficients of x, y and z in the numerators must be 1	**Need:** 1 point + 1 direction or 2 points A, B: $\mathbf{b}=\overrightarrow{AB}$
Plane	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ $\mathbf{r}\cdot\mathbf{n}=d$ (scalar product form)	$n_x x+n_y y+n_z z=d$ $\mathbf{n}=\begin{pmatrix}n_x\\n_y\\n_z\end{pmatrix}$	**Need:** 1 point + 2 non-parallel directions // plane or 1 point + normal $\mathbf{n}$ $\mathbf{n}=\mathbf{b}\times\mathbf{c}$, $d=\mathbf{a}\cdot\mathbf{n}$ 3 points: $\mathbf{b}=\overrightarrow{AB}$, $\mathbf{c}=\overrightarrow{AC}$; contains a line: use its direction

$\mathbf{a}$: position vector of any point on the line/plane. $\mathbf{b}$, $\mathbf{c}$: any two linearly independent directions parallel to the plane.

The equation of any line is not unique.

Note

To check if a given point C lies on
- a line: $\overrightarrow{OC}=\mathbf{a}+\lambda\mathbf{b}$ (one λ satisfies all 3 components)
- a plane: $\overrightarrow{OC}\cdot\mathbf{n}=d$

Example 2.1

Line through $(1,2,3)$, $(3,2,2)$

$$\mathbf{b}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$\frac{x-1}{2}=\frac{z-3}{-1},\quad y=2$$

Example 2.2

Plane through $B(0,1,1)$, $C(1,1,3)$, $D(2,0,2)$

$$\mathbf{n}=\overrightarrow{BC}\times\overrightarrow{BD}=\begin{pmatrix}1\\0\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\3\\-1\end{pmatrix}$$

$$d=\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\-1\end{pmatrix}=2$$

$$2x+3y-z=2$$

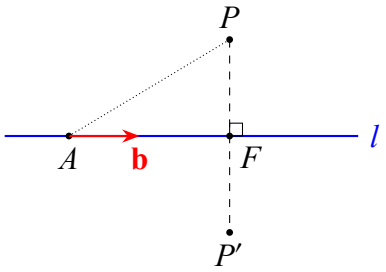
2.2 Interactions Map

	Point	Line	Plane
Point	distance $\lvert\overrightarrow{AB}\rvert$		
Line	• on line? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / intersect / skew §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in line §2.7	
Plane	• on plane? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / lies in / intersect §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in plane §2.7	• // / same / intersect §2.4 • line of intersection §2.4 • angle §2.6 • distance (//) §2.8

Always draw the diagram first.

2.3 Foot / Distance / Reflection

Point P – Line $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$

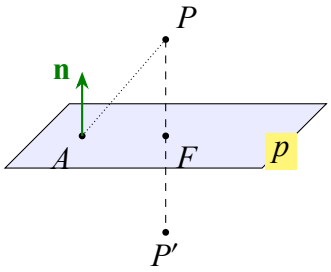


1. $\overrightarrow{OF}=\mathbf{a}+\lambda\mathbf{b}$ for some $\lambda\in\mathbb{R}$
2. $\overrightarrow{PF}\cdot\mathbf{b}=0 \Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

$$PF=\lvert\overrightarrow{AP}\times\hat{\mathbf{b}}\rvert$$

Point P – Plane $\mathbf{r}\cdot\mathbf{n}=d$



1. $l_{PF}:\ \mathbf{r}=\overrightarrow{OP}+\lambda\mathbf{n},\ \lambda\in\mathbb{R}$
2. $(\overrightarrow{OP}+\lambda\mathbf{n})\cdot\mathbf{n}=d$ (intersection between line & plane) $\Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

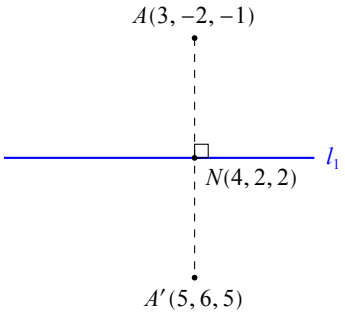
$$PF=\lvert\overrightarrow{AP}\cdot\hat{\mathbf{n}}\rvert$$

Reflection (and foot of perpendicular) of point to line

Example 2.3

Find the foot of the perpendicular N from $A(3,-2,-1)$ to the line l_1, and the reflection A' of A in l_1, where

$$l_1: \mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}, \quad \lambda\in\mathbb{R}$$



$$\overrightarrow{AN}=\begin{pmatrix}-2+3\lambda\\4\\4-\lambda\end{pmatrix}$$

$$\overrightarrow{AN}\cdot\begin{pmatrix}3\\0\\-1\end{pmatrix}=0$$

$$10\lambda-10=0 \Rightarrow \lambda=1$$

$$\therefore N=(4,2,2)$$

$$\overrightarrow{ON}=\frac{\overrightarrow{OA}+\overrightarrow{OA'}}{2} \quad \text{(midpoint/ratio theorem)}$$

$$\Rightarrow \overrightarrow{OA'}=2\overrightarrow{ON}-\overrightarrow{OA}$$

$$\overrightarrow{OA'}=2\begin{pmatrix}4\\2\\2\end{pmatrix}-\begin{pmatrix}3\\-2\\-1\end{pmatrix}$$

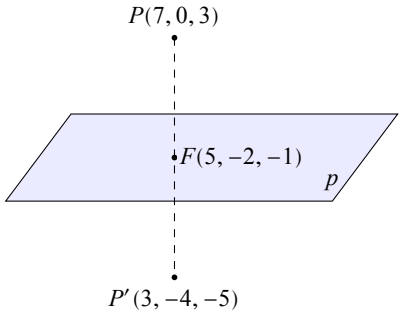
$$\therefore A'=(5,6,5)$$

Reflection (and foot of perpendicular) of point to plane

Example 2.4

Find the foot of the perpendicular F from $P(7,0,3)$ to the plane p, and the reflection P' of P in p, where

$$p: \mathbf{r}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$



$$l_{PF}: \mathbf{r}=\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

$$\left(\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$

$$13+6\lambda=1 \Rightarrow \lambda=-2$$

$$\therefore F=(5,-2,-1)$$

$$\overrightarrow{OF}=\frac{\overrightarrow{OP}+\overrightarrow{OP'}}{2} \quad \text{(midpoint/ratio theorem)}$$

$$\Rightarrow \overrightarrow{OP'}=2\overrightarrow{OF}-\overrightarrow{OP}$$

$$\overrightarrow{OP'}=2\begin{pmatrix}5\\-2\\-1\end{pmatrix}-\begin{pmatrix}7\\0\\3\end{pmatrix}$$

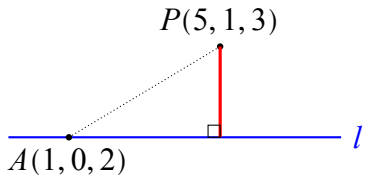
$$\therefore P'=(3,-4,-5)$$

Shortest distance from point to line

Example 2.5

Find the shortest distance from $P(5,1,3)$ to the line

$$l: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$



$$\text{shortest distance}=|\overrightarrow{AP}\times\hat{\mathbf{b}}|$$

$$=\left|\begin{pmatrix}4\\1\\1\end{pmatrix}\times\frac{1}{3}\begin{pmatrix}2\\-1\\2\end{pmatrix}\right|$$

$$=\left|\frac{1}{3}\begin{pmatrix}3\\-6\\-6\end{pmatrix}\right|=\left|\begin{pmatrix}1\\-2\\-2\end{pmatrix}\right|$$

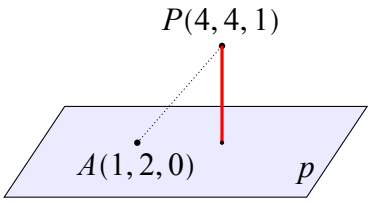
$$=3$$

Shortest distance from point to plane

Example 2.6

Find the shortest distance from $P(4,4,1)$ to the plane

$$p: \mathbf{r}\cdot\begin{pmatrix}2\\1\\-2\end{pmatrix}=4$$



$$\text{shortest distance}=|\overrightarrow{AP}\cdot\hat{\mathbf{n}}|$$

$$=\left|\begin{pmatrix}3\\2\\1\end{pmatrix}\cdot\frac{1}{3}\begin{pmatrix}2\\1\\-2\end{pmatrix}\right|$$

$$=\left|\frac{1}{3}(6+2-2)\right|$$

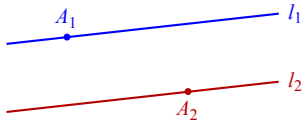
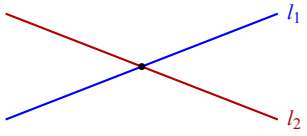
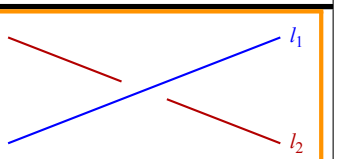
$$=2$$

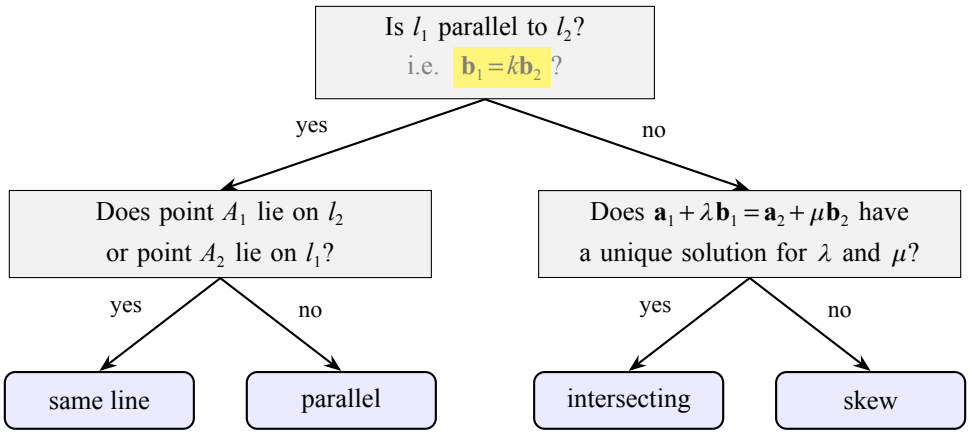
Note

We can also find the coordinates of the foot of the perpendicular N first, then take $|\overrightarrow{PN}|$, but this is less efficient.

2.4 Relationships and Intersections

Line – Line $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1$, $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2$

	Parallel	Intersecting	Skew
Diagram	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$
Condition	$\mathbf{b}_1 \parallel \mathbf{b}_2$ and $A_1 \notin l_2, A_2 \notin l_1$	$\mathbf{b}_1 \nparallel \mathbf{b}_2$, unique solution for λ and μ	$\mathbf{b}_1 \nparallel \mathbf{b}_2$, no solution for λ and μ



Note

Definition of **skew lines**:

- non parallel
- non intersecting

Thus, to prove that two lines are skew, we must first state that $\mathbf{b}_1 \neq k\mathbf{b}_2$, so l_1 and l_2 are not parallel, before proceeding with simultaneous equations to show they have no solution.

Example 2.7

Determine whether l_1 and l_2 are intersecting or skew lines.

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 12 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}, \quad \mu \in \mathbb{R}$$

$$1 + \lambda = 2 - \mu \qquad \text{—(1)}$$

$$2\lambda = 12 + 3\mu \qquad \text{—(2)}$$

$$2 + \lambda = 5 \qquad \text{—(3)}$$

From (1) and (3), $\lambda = 3, \quad \mu = -2$

Sub into (2), LHS $= 6$, RHS $= 12 + 3(-2) = 6$

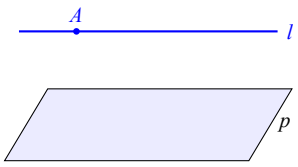
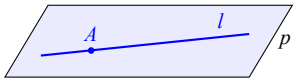
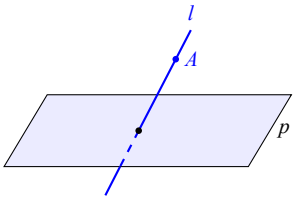
$\therefore$ LHS $=$ RHS

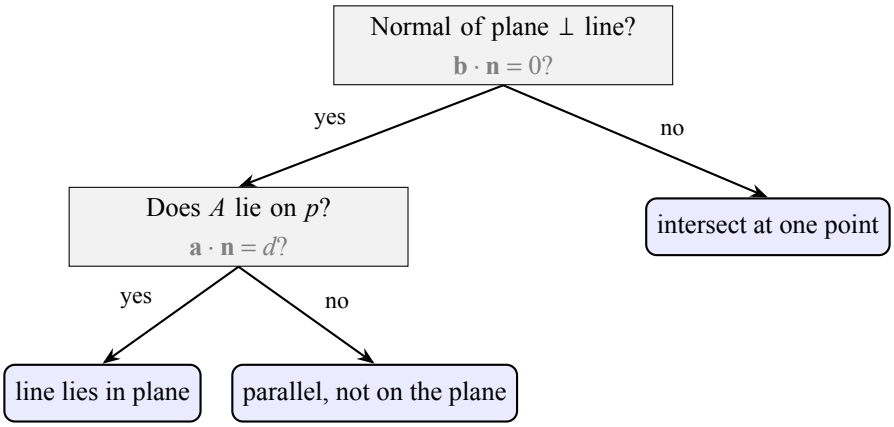
Sub $\lambda = 3$ into $l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix}$

They intersect at $(4, 6, 5)$

Solve 2 of the 3 equations, sub into the 3rd: LHS = RHS ⇒ intersect; LHS ≠ RHS ⇒ skew.

Line – Plane $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$, $p: \mathbf{r}\cdot\mathbf{n}=d$

	Parallel, not on the plane	Line lies in the plane	Intersects at one point
Diagram	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$
Condition	• normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A does not lie on p: $\mathbf{a}\cdot\mathbf{n}\neq d$	• normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A lies on p: $\mathbf{a}\cdot\mathbf{n}=d$	• normal not $\perp$ line: $\mathbf{b}\cdot\mathbf{n}\neq 0$



Example 2.8

Find the point of intersection between the line l and plane p.

$$l:\ \mathbf{r}=\begin{pmatrix}1\\3\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$p:\ \mathbf{r}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5$$

$$\mathbf{r}=\begin{pmatrix}1\\3\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\1\end{pmatrix}=\begin{pmatrix}1+\lambda\\3-2\lambda\\1+\lambda\end{pmatrix},\quad \lambda\in\mathbb{R}$$

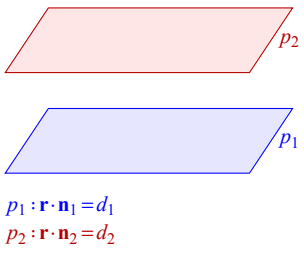
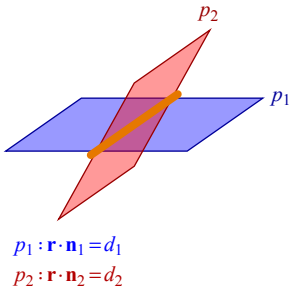
$$\begin{pmatrix}1+\lambda\\3-2\lambda\\1+\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5$$

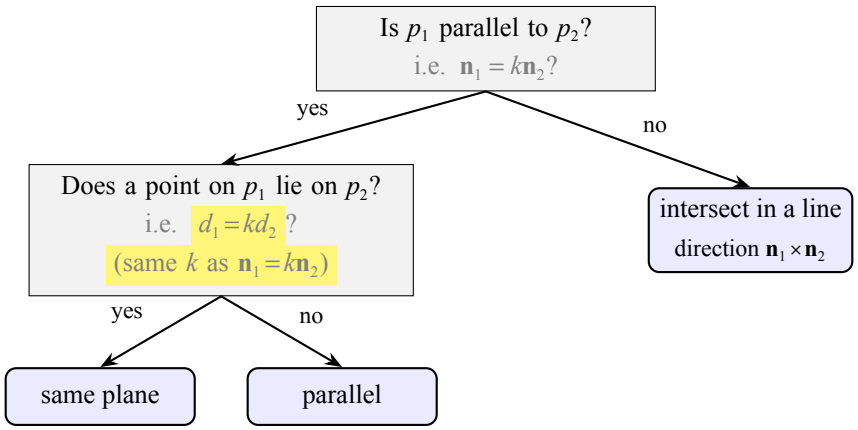
$$8-2\lambda=5\ \Rightarrow\ \lambda=\tfrac{3}{2}$$

$$\therefore\ (2.5,0,2.5)$$

Plane – Plane $p_1 : \mathbf{r}\cdot\mathbf{n}_1 = d_1,\ p_2 : \mathbf{r}\cdot\mathbf{n}_2 = d_2$

Let $p_1 :\ \mathbf{r}\cdot\mathbf{n}_1 = d_1$ and $p_2 :\ \mathbf{r}\cdot\mathbf{n}_2 = d_2$ be distinct planes.

	Parallel	Intersecting in a line
Diagram		
Condition	$\mathbf{n}_1 \parallel \mathbf{n}_2, p_1 \neq p_2$	$\mathbf{n}_1 \nparallel \mathbf{n}_2$



Example 2.9

Find the intersection between the planes:

$$3x+2y+z=10$$

$$x-2y+z=5$$

GC: $x=\frac{15}{4}-\frac{1}{2}z$

$y=-\frac{5}{8}+\frac{1}{4}z$

$z=z$

$$\therefore l:\ \mathbf{r}=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+\lambda\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

Example 2.10

Sometimes, we may not be able to use our GC to find the line of intersection between two planes. The steps to find it manually are:

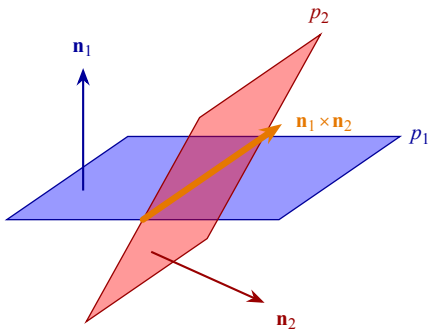
$$p_1: \mathbf{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=6, \qquad p_2: \mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=2$$

Step 1: direction $=\mathbf{n}_1\times\mathbf{n}_2$

$$\mathbf{b}=\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$=\begin{pmatrix}1(2)-1(-1)\\1(1)-1(2)\\1(-1)-1(1)\end{pmatrix}$$

$$=\begin{pmatrix}3\\-1\\-2\end{pmatrix}$$



Step 2: find a point that lies on both planes

$$p_1: x+y+z=6$$

$$p_2: x-y+2z=2$$

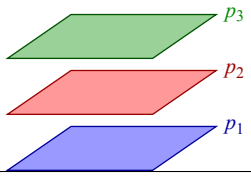
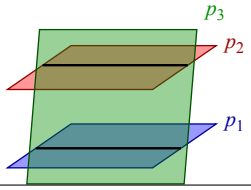
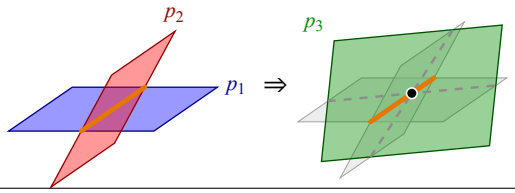
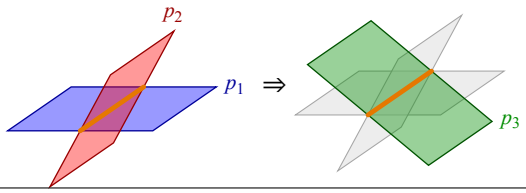
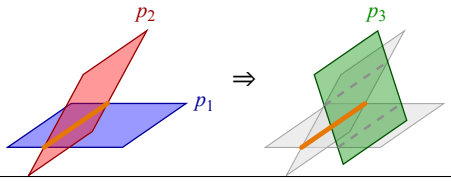
$$\text{Let } z=0$$

$$\left.\begin{matrix}x+y=6\\x-y=2\end{matrix}\right\}\Rightarrow\begin{matrix}x=4\\y=2\end{matrix}$$

$$\therefore l: \mathbf{r}=\begin{pmatrix}4\\2\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\-1\\-2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

2.5 Interaction between 3 Planes

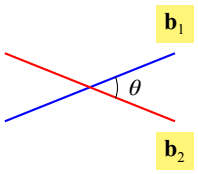
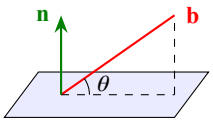
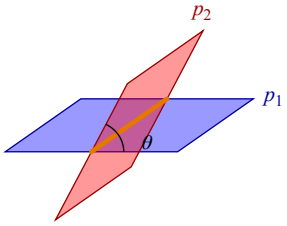
Let $p_1: \mathbf{r}\cdot\mathbf{n}_1 = d_1$, $p_2: \mathbf{r}\cdot\mathbf{n}_2 = d_2$ and $p_3: \mathbf{r}\cdot\mathbf{n}_3 = d_3$ be distinct planes.

Case	Diagram(s)	Condition l: line of intersection of p_1 and p_2 ($\mathbf{n}_1 \nparallel \mathbf{n}_2$)
1. All parallel		$\mathbf{n}_1 \parallel \mathbf{n}_2 \parallel \mathbf{n}_3$
2. Two parallel, one not		$\mathbf{n}_1 \parallel \mathbf{n}_2$, but $\mathbf{n}_3$ not parallel to them
3. All intersect at 1 point (most common)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ intersects p_3 (see p. 9)
4. All intersect in a line		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ lies on p_3 (see p. 9)
5. Form a triangular prism (no single point of intersection)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ • $l \,//\, p_3$: $\mathbf{b}\cdot\mathbf{n}_3 = 0$ • $l \not\subset p_3$: $\mathbf{a}\cdot\mathbf{n}_3 \neq d_3$ (see p. 9)

Note

i To analyse the interaction between 3 arbitrary planes, we can study how the **line of intersection between p_1 and p_2** interacts with p_3.

2.6 Angles

Line – Line	Line – Plane	Plane – Plane
		
$\cos\theta = \dfrac{\lvert\mathbf{b}_1\cdot\mathbf{b}_2\rvert}{\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert}$	$\sin\theta = \left\lvert\dfrac{\mathbf{n}\cdot\mathbf{b}}{\lvert\mathbf{n}\rvert\lvert\mathbf{b}\rvert}\right\rvert$	$\cos\theta = \left\lvert\dfrac{\mathbf{n}_1\cdot\mathbf{n}_2}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}\right\rvert$
directions	sin: $\angle(\mathbf{n},\mathbf{b}) = 90^\circ - \theta$	normals

θ = acute angle.

Each formula has a modulus in the numerator $\because \theta$ must be acute.

Example 2.11

Find the angle between the planes $3x-y+2z=3$ and $y=2$.

$$\cos\theta = \frac{|-1|}{\sqrt{14}\cdot 1}$$
$$\theta = 74.5^\circ$$

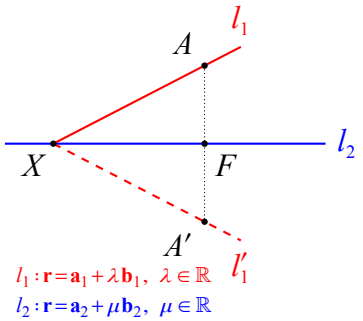
Example 2.12

Find the angle between the line l and plane p.

$$l:\ \mathbf{r} = \begin{pmatrix}-2\\-4\\5\end{pmatrix} + \lambda\begin{pmatrix}3\\5\\-4\end{pmatrix},\quad \lambda\in\mathbb{R}$$
$$p:\ \mathbf{r}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix} = 1$$
$$\sin\theta = \frac{5}{\sqrt{50}\cdot 1} = \frac{1}{\sqrt{2}}$$
$$\theta = 45^\circ$$

2.7 Reflection of a Line

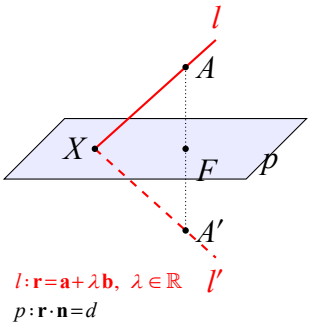
Line l_1 in line l_2 (intersect at X)



1. $X=l_1\cap l_2$
2. $A\in l_1$: foot F on l_2
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l_1': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

If $l\ //\ p$: l' has the same direction, through A'.

Line l in plane p (intersect at X)



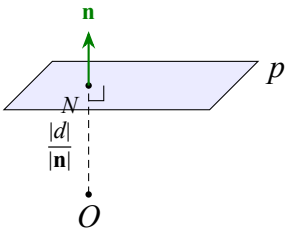
1. $X=l\cap p$
2. $A\in l$: foot F on p
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

2.8 Distance Between Parallel Planes

$$p:\ \mathbf{r}\cdot\mathbf{n}=d$$

Distance of plane from origin

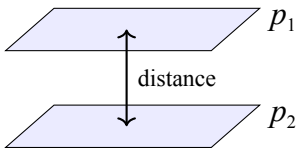
$$\text{distance from } O=\frac{|d|}{|\mathbf{n}|}$$



$$p_1:\ \mathbf{r}\cdot\mathbf{n}=d_1,\qquad p_2:\ \mathbf{r}\cdot\mathbf{n}=d_2$$

Distance

$$\text{distance}=\left|\frac{d_1}{|\mathbf{n}|}-\frac{d_2}{|\mathbf{n}|}\right|$$



Same $\mathbf{n}$ for both planes first. $\mathbf{r}\cdot\mathbf{n}=0$: plane through O.

Example 2.13

Find the distance between the two planes $x+2y-3z=2$ and $2x+4y-6z=-10$.

$$2x+4y-6z=-10\ \Rightarrow\ x+2y-3z=-5$$

$$\text{distance}=\left|\frac{2}{\sqrt{14}}-\frac{-5}{\sqrt{14}}\right|=\frac{7}{\sqrt{14}}=\frac{\sqrt{14}}{2}$$